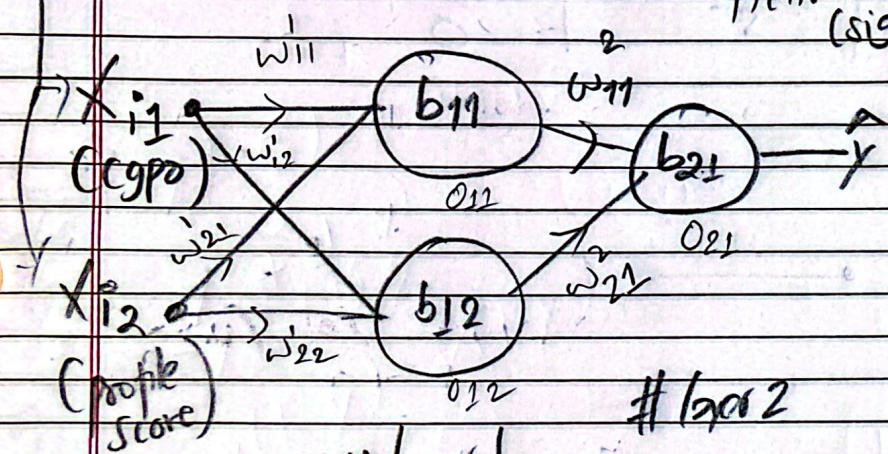


Classification Example

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cgpa	profile score	placement
8	8	1
7	9	1
6	10	0
5	5	0

Activation function (sigmoid)



layer 0
layer 1
layer 2
 $2 \times 2 + 2 = 6$
 $2 \times 1 + 1 = 3$

Total 9 trainable parameters

Binary cross entropy loss function

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$$Loss = -y \log(\hat{y}) - (1-y) \log(1-\hat{y})$$

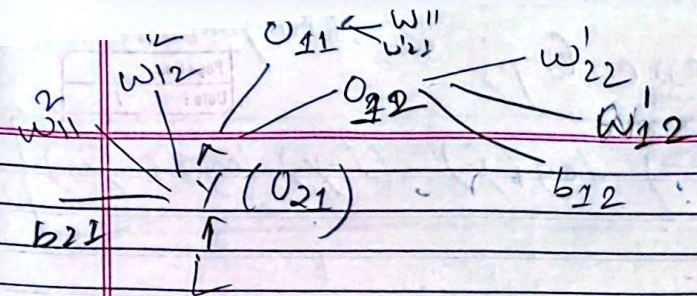
Back propagation Algorithm

epochs = 5

for i in range(epochs):

for j in range(x.shape[0]):

- select 1 row (random or $w \rightarrow 1, b \rightarrow 0$)
- predict (using forward propagation)
- calculate loss (using loss function $\rightarrow BCE/CE$)
- update weights & bias using Gradient Descent
- $W_1 = W_0 - \eta \frac{\partial L}{\partial W}$
- Calculate average loss for the epoch



Derivative wrt loss to all 9 trainable parameters.

$$1) \frac{\partial L}{\partial w_{11}^2} = \left(\frac{\partial L}{\partial \hat{y}} \right) * \left(\frac{\partial \hat{y}}{\partial w_{11}^2} \right)$$

$$\begin{aligned} & \frac{\partial}{\partial \hat{y}} (-y \log(\hat{y}) - (1-y) \log(1-\hat{y})) \\ &= -\frac{y}{\hat{y}} + \frac{(1-y)}{1-\hat{y}} \\ &= -\frac{y(1-\hat{y}) + \hat{y}(1-y)}{\hat{y}(1-\hat{y})} \\ &= -\frac{-y + y\hat{y} + \hat{y} - y\hat{y}}{\hat{y}(1-\hat{y})} \\ &= -\frac{\hat{y} - y}{\hat{y}(1-\hat{y})} \end{aligned}$$

Remembers this, very important

$$\rightarrow \frac{\partial (\sigma(z))}{\partial z} = \sigma(z) [1 - \sigma(z)]$$

$$\rightarrow = \frac{\partial \hat{y}}{\partial w_{11}^2} = \frac{\partial \sigma(z)}{\partial z} (w_{11}^2 + w_{12}^2 + w_{21}^2 + w_{22}^2 + b_{11} + b_{12})$$

$$\boxed{\hat{y} = \sigma(z)} \quad \frac{\partial \sigma(z)}{\partial z} = \sigma(z) [1 - \sigma(z)]$$

Remember

$$= \hat{y} (1 - \hat{y})$$

So,

$$\frac{\partial L}{\partial w_{11}^2} = \frac{-(y - \hat{y}) * \hat{y} (1 - \hat{y})}{\hat{y} (1 - \hat{y})}$$

$$= -(y - \hat{y})$$

So on (Interodtly same as regression)